

Review Manipulation and Filtering on Digital Platforms

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Abstract. Digital platforms strive to filter out consumer reviews that are manipulated, which has become a common and increasingly vexing problem that can take anywhere from days to months to detect. Yet little is known about the consequences of such reviews on product market performance. Using data from the Apple App Store, we examine how the app's ranking changes when it receives manipulated reviews that are later filtered out by Apple. Our findings reveal that both one-star and five-star manipulated reviews have a significant positive effect on app rankings within a week of posting. The positive effect of one-star manipulated reviews is particularly surprising, as it contradicts the expected effect of organic negative reviews and the intent behind using such reviews to harm competitors. We also explore how these effects evolve over time, shedding light on the role of filtering policies in mitigating distortions related to review manipulation. Results show that these effects become negative as platforms filter out manipulated reviews, but this process can take as long as six months to fully materialize. These findings highlight the need for digital platforms to increase their investment to promptly and accurately control review manipulation for the welfare of both businesses and users on the platform. To the best of our knowledge, this study is one of the first to empirically analyze the short-term and long-term effects of manipulated reviews on product sales, which provides crucial managerial implications for practitioners.

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1. Introduction

Online consumer reviews have long been recognized as an important factor in consumers' purchase decisions. A substantial body of research has shown that online consumer reviews affect sales and revenues of various products and services such as television viewership (Godes and Mayzlin 2004), movies' box office revenue (Liu 2006), book sales (Chevalier and Mayzlin 2006), and digital camera sales (Archak et al. 2011). Although the importance of online consumer reviews has been widely acknowledged, the value of reviews has also been threatened by the common and increasingly vexing problem of review manipulation.

According to the review policies of major digital platforms such as Google, Yelp, and Airbnb,¹ most platforms prohibit manipulated reviews such as reviews written by unethical businesses to promote their own

products or harm the reputations of their competitors' products. They also prohibit businesses from incentivizing consumers to write positive reviews, another prevalent form of review manipulation. Although consumers expect to learn useful information from online reviews, manipulated reviews can be detrimental to users, businesses, and digital platforms. For instance, manipulated reviews are often unfairly biased and could mislead consumers into disadvantageous purchases. Furthermore, negative manipulated reviews may cost targeted businesses substantial efforts and resources to mitigate damage. Lack of rigorous oversight over manipulated reviews undoubtedly results in users and businesses abusing such a system, wherein consumer reviews will lose much of their usefulness and credibility when the platform is flooded with manipulated reviews.

To retain consumer trust and provide businesses with a healthy marketplace, digital platforms such as Amazon, the Apple App Store, Google, and Yelp endeavor to filter out (identify and remove) consumer reviews that violate platform policies. In addition to manipulated reviews, platforms also filter out reviews with useless content that does not address consumer experiences, such as those containing advertising and solicitation, gibberish, or a person's political, social, or religious commentary. Reviews with content that is offensive and regulated, such as harassment, hate speech, violence, and illegal content, are also filtered out.² Nevertheless, the platforms' primary goal in filtering reviews is to mitigate review manipulation, as supported by our empirical evidence based on text analysis that 88% of filtered reviews are manipulated reviews in our data from the Apple App Store.

There are two popular review filtering strategies platforms apply: "prefilter," in which consumer reviews are checked for policy compliance before becoming visible to other platform users, and "postfilter," wherein policy-violating reviews are removed after they have been posted and seen by other users (Leetaru 2019). The prefilter strategy generally ensures that the content quality of reviews is well-maintained and that undesired content is kept out of the visible community. This strategy nevertheless deprives the review writers of immediate gratification because they must wait for an unpredictable length of time for their submission to be approved by the platform, which is often seen as an important cause of online community failure (Grimes-Viort 2010). Therefore, major platforms usually adopt the postfilter strategy and utilize an automated filter that combines machine learning and artificial intelligence to discover policy-violating reviews at scale, but they also rely on human moderation and reports from businesses and users (Wiggers 2020). For example, 25% of consumer reviews on Yelp are identified as policy-violating and are thus filtered after posting (Ha 2021). Google also removed 75 million posted reviews from Google Maps in 2019 (Wiggers 2020).

As it can take anywhere from days to months for the platforms to remove reviews, a huge number of manipulated reviews may reach millions of users before being filtered by the platforms (Nguyen 2018, Leetaru 2019, Pierce et al. 2021). Yet little is known about the consequences, especially the sales outcomes, of such manipulated reviews before they are filtered by the platform. To address this gap, this study aims to quantify how a product's sales change when it receives manipulated reviews. We also investigate how the effects of manipulated reviews change over time. This is especially important for understanding the role of filtering policies in mitigating distortions related to review manipulation. Overall, our research provides insights into the justifiability of platform

actions in filtering manipulated reviews and explores potential unintended effects. Our findings are particularly relevant for platforms, policymakers, and researchers navigating the complex intersection of user trust, content quality, and the integrity of online information dissemination.

We analyze a unique data set containing daily app rankings and reviews posted in the Apple App Store over a one-year period from 2018 through 2019. We observe not only the reviews that are always visible to App Store users but also those that are filtered by Apple after posting and thus become invisible to App Store users. Our study shows that both one-star and five-star manipulated reviews have a significant positive effect on app rankings within a week of posting. This is particularly surprising for one-star manipulated reviews, as they are typically expected to lower an app's ranking if they are perceived as organic negative reviews. Instead, these negative manipulated reviews, often used by unethical businesses to attack competitors, paradoxically enhance an app's visibility. Five-star manipulated reviews, as expected, have a positive effect on app rankings, reinforcing the distortion in competitive dynamics on digital platforms. In addition, we show that manipulated reviews have a greater effect on free apps, gaming apps, and apps from large developers. Also, reviews with useless and offensive content both show significant positive effects on app rankings, with offensive content having larger effects. These findings reinforce our main results and highlight the importance of review volume over valence, particularly for negative reviews.

Moreover, the effects of manipulated reviews evolve over time. Although platforms like the App Store implement filtering mechanisms, the initial positive impact of these reviews gradually diminishes and can eventually turn negative. This indicates that the distortion caused by manipulated reviews is slowly corrected by the platform's filtering mechanisms. But as this correction process is slow, taking as long as six months to fully take effect, the influence of manipulated reviews can persist longer than expected and is not easy to neutralize. Our findings underscore how crucial effective filtering policies are in mitigating these distortions, highlighting the need for digital platforms to invest in timely and robust detection strategies to maintain fair competition and marketplace integrity.

This study makes several important contributions to the literature on review manipulation by uncovering the downstream consequences of manipulated online reviews on product market outcomes, particularly in terms of app sales rankings. Using a unique data set of app reviews that were later filtered by the platform, we provide empirical evidence that both incentivized reviews posted by the focal app and

negative reviews posted by competitors are positively associated with app sales rankings. By doing so, our study is among the first to reveal that review manipulation—whether self-promotional or aimed at damaging rivals—can, under certain conditions, enhance market performance rather than harm it. Beyond immediate effects, we also explore the long-term implications of review manipulation, offering evidence that platform-level filtering strategies may help mitigate potential harms over time. Methodologically, we advance the literature by employing a large language model to better differentiate manipulated reviews from other filtered content—such as irrelevant or offensive comments—enabling a more precise identification of manipulation in observational data.

We further contribute by proposing a conceptual framework that discusses the underlying mechanisms behind the observed effects, drawing on the well-established volume and valence framework from the organic online review literature. Specifically, we demonstrate that the impact of manipulated reviews on consumer behavior depends on whether consumers recognize the reviews as manipulated. Notably, our findings uncover a possible asymmetry in consumer detection—users may be more likely to identify negative manipulated reviews, whereas positive manipulated reviews may be overlooked and perceived as organic. We provide supporting evidence for this pattern through a text analysis, showing that negative manipulated reviews are linguistically more distinct from organic reviews, making them easier to detect. In addition, we explore heterogeneity in the effects of review manipulation by different app characteristics, offering further support for our proposed mechanisms.

Our study also provides important managerial implications for platform managers and firms. We find that manipulated reviews have positive effects on sales rankings, but rankings drop slowly as these reviews are removed. This highlights the powerful role of misinformation in shaping consumer decisions and the need for platforms to accelerate and improve their filtering processes. Faster intervention would help reduce misinformation and prevent consumers from making suboptimal purchasing decisions. Moreover, although firms often worry that negative manipulated reviews posted by competitors will damage their reputation, our study shows a positive effect, which suggests that attempts to damage competitors through negative reviews may not have their intended outcome.

2. Literature Background

Our research objective is to investigate the effects of review manipulation on product sales. In this section, we review the existing literature on review manipulation,

the valence and volume effects of online reviews, and platform governance.

2.1. Review Manipulation

Online reviews play a critical role in shaping consumer decisions and business outcomes, prompting some firms to engage in review manipulation—deliberately altering review content to mislead consumers through product misrepresentation (Ansari and Gupta 2021). Such manipulation can take two primary forms: positive manipulation, where firms promote their own products, and negative manipulation, where firms target competitors to damage their reputations (Nie et al. 2022).

In the case of positive manipulation, firms typically post artificially favorable reviews to boost their image and attract more customers, particularly in response to intense competition or reputational pressure (Chevalier and Mayzlin 2006, Dellarocas 2006, Mayzlin 2006). Conversely, negative manipulation involves posting fraudulent or critical reviews aimed at reducing a competitor's ratings and eroding consumer trust, ultimately to divert market share (Lappas 2012, Mayzlin et al. 2014, Luca and Zervas 2016, Nie et al. 2022).

Prior research has primarily focused on uncovering the motivations and strategies of firms for review manipulation. Such practices are more prevalent in highly competitive markets and among firms with weaker reputations or lower-quality products (Mayzlin 2006, Mayzlin et al. 2014, Luca and Zervas 2016). For instance, Nie et al. (2022) showed that hotels facing new competitors shift their strategy away from undermining rivals and instead invest more in self-promotional review tactics. Beyond explicit manipulation, prior studies also examine how firms stimulate review generation through product sampling (Mo and Li 2018, Lin et al. 2019), cashback incentives (Wu et al. 2023), and solicitation efforts (Gao et al. 2024), which can crowd out organic reviews and distort consumer perceptions. Yet, although prior work has shed light on the motivations and tactics behind review manipulation, much less is known about the consequences on product sales before the reviews are detected and removed by the platform. We propose a theoretical framework to explain the potential effects of manipulated reviews on app sales ranking, bridging the existing literature on both organic and manipulated reviews. We also conduct a text analysis to compare the content of organic and manipulated reviews.

Although some studies, such as Li et al. (2022), suggest that earlier product reviews can have a lasting impact on subsequent product releases, much of the existing research on manipulated reviews has concentrated on their short-term effects, and the longer-term consequences of review manipulation on market outcomes remain unclear. However, understanding the

temporal dynamics of these effects is crucial, as the initial boost in visibility or ranking may not translate into sustained market success. If manipulated reviews generate only temporary gains or eventually backfire because of detection or removal, then their net impact may change substantially over time. Our study addresses this gap by evaluating the impact of both positive and negative manipulated reviews on product sales across a longer horizon. In doing so, we aim to analyze whether, why, and when manipulated reviews may help or hurt the sales outcomes of a product.

2.2. Valence and Volume Effects of Online Review

Previous research has recognized valence and volume as two key forces through which organic reviews influence consumer behavior.

2.2.1. Valence Effects. Valence refers to the sentiment expressed in reviews, which can range from positive to negative or neutral (You et al. 2015). Numerous studies have shown that review valence exerts a persuasive influence on consumers' perceptions of product quality and their overall evaluation of a product (Duan et al. 2008, Jabr and Rahman 2022). Positive reviews tend to reinforce expectations, generate favorable impressions, and encourage purchase decisions (Fang 2022, Zhao et al. 2022). In contrast, negative reviews often signal performance issues, prompting skepticism and lowering purchase intent (Liu 2006). When consumer opinions are already mixed, additional negative valence can amplify uncertainty and cause delays in purchase decisions (Lee et al. 2023, Yin et al. 2023).

2.2.2. Volume Effect. Volume refers to the total number of conversations in which consumers engage about a particular product (You et al. 2015). Unlike valence, which affects perceptions of product quality, volume primarily works through an awareness mechanism: the more reviews a product has, the more visible it becomes, which can lead to greater consumer exposure and higher sales (Godes and Mayzlin 2004, Liu 2006, Duan et al. 2008).

2.2.3. Relative Importance of Valence and Volume Effects. Both valence and volume of reviews influence consumer behavior, and previous research has shown that these two factors can jointly shape their overall impact (Chen et al. 2008, Khare et al. 2011). Nevertheless, their relative importance has been debated. Floyd et al. (2014) suggested that average product ratings have a greater influence than the number of reviews, whereas You et al. (2015) argued that the volume effect is more critical. The examination by Watson

et al. (2018) of the relative importance of valence and volume effects showed that although average ratings are a more useful indicator of product quality, the number of reviews becomes more impactful when the average ratings are already high.

Although valence and volume effects are well-documented for organic reviews, their influence in the context of manipulated reviews remains relatively underexplored. Crucially, manipulated reviews may not exert the same impact as organic ones because of key differences in their intent and perceived authenticity. Organic reviews are typically generated by users based on genuine consumption experiences, with the intent to inform or share opinions with other consumers. In contrast, manipulated reviews are often created strategically—either to promote a product or to discredit a competitor—with the primary goal of shaping perception rather than reflecting honest feedback. This divergence in intent can affect how consumers interpret and react to reviews. Moreover, manipulated reviews may vary widely in perceived authenticity: some are carefully crafted to mimic high-quality, believable content, whereas others are vague, exaggerated, or transparently fake. When consumers suspect or recognize that reviews are inauthentic, their persuasive power may diminish or even backfire, triggering skepticism or distrust (Zhuang et al. 2018, Harrison-Walker and Jiang 2023). Our study aims to connect prior research on the valence and volume effects of organic reviews with the emerging literature on manipulated reviews by investigating how these effects differ in the context of review manipulation. Specifically, we examine how variations in consumer information processing and perceived authenticity between organic and manipulated reviews influence their impact on product sales.

2.3. Platform Governance

In response to the growing prevalence of review manipulation, a stream of research has investigated how digital platforms attempt to mitigate such behavior through governance mechanisms. For instance, Chen et al. (2022) proposed that conditional rebate strategies on e-commerce platforms could incentivize manipulated reviews when the cost of posting reviews is low. Similarly, Kokkodis et al. (2022) suggested that optional purchase verification could reduce manipulated reviews by introducing indirect costs. Beck et al. (2024) highlighted that monitoring online review practices could be a key governance tool to mitigate review manipulation and build customer trust. Although much research on review filtering has emphasized its impact on customer trust, Jiang et al. (2023) found that transparency in review filtering may discourage users from contributing reviews. In contrast, Ananthakrishnan et al. (2020) proposed that displaying filtered

reviews could enhance customer trust in the platform's review information. Despite this, the impact of review filtering has received relatively little attention. Although it is designed to detect manipulated reviews, its broader effects, both before and after implementation, are underexplored.

Identifying manipulated reviews is crucial for both platform managers and researchers. But this task presents an empirical challenge because manipulated reviews—designed to mimic real reviews by consumers—are not directly observable, making it difficult to distinguish between genuine and falsified feedback (Mayzlin et al. 2014, Lappas et al. 2016). Some studies have compared platforms like TripAdvisor, which allows any user to post reviews, with Expedia, which restricts reviews to verified customers (Mayzlin et al. 2014, Nie et al. 2022). Other studies have used simulations to analyze the impact of manipulated reviews on platforms (Lappas et al. 2016). More recently, researchers have also used platform-filtered reviews as a proxy for identifying manipulated reviews (Luca and Zervas 2016, Kokkodis et al. 2022). However, filtered reviews may also contain reviews with irrelevant or offensive content that violate platform policies and are removed by the platform. Our study employs platform-filtered reviews and a large language model to better distinguish manipulated reviews from other filtered reviews, providing a more accurate measure of these manipulated reviews based on the actual filtered reviews.

Our study extends the literature on online reviews and platform governance in three ways. First, we extend prior work by empirically examining the market impact of manipulated reviews, shedding light on how both positive and negative manipulation can influence product sales—sometimes in unexpected ways. We further explore how these effects evolve over time, offering a dynamic perspective on manipulation and filtering. Second, we contribute methodologically by proposing a novel approach to identifying manipulated reviews, combining platform-filtered data with a large language model to improve detection accuracy and differentiate manipulation from other forms of policy-violating content. Third, we propose a theoretical extension that connects the valence and volume effects known in organic reviews to the manipulated review context, providing insights on how these effects might be altered when consumer awareness of review manipulation is different. To highlight the contributions of our research, we present an overview of the relevant literature on review manipulation in Table 1.

3. Theoretical Framework

3.1. Conceptual Background

Although the existing literature on online reviews has largely focused on organic, user-generated content,

manipulated reviews introduce a distinct layer of complexity. Our conceptual framework begins by acknowledging that manipulated reviews can exhibit both similarities and differences compared with their organic counterparts—particularly in terms of volume effects, valence effects, and how these are shaped by consumers' awareness of the manipulated reviews.

First, manipulated reviews are likely to operate similarly to organic reviews in terms of volume effects. Volume—referring to the total number of reviews—plays a key role in increasing a product's visibility and perceived popularity, often functioning as a shortcut for product awareness and credibility (Duan et al. 2008, Luca and Zervas 2016). Many consumers focus on the number of reviews rather than their content when making purchase decisions. As a result, the presence of manipulated reviews, regardless of their authenticity, can inflate the total number of visible reviews and thereby enhance product exposure. In this regard, we expect the volume effect of manipulated reviews to mirror that of organic reviews.

However, valence effects, which refer to the sentiment of reviews, may differ significantly between manipulated and organic reviews, depending on whether consumers are able to detect the manipulation. When manipulated reviews are well-crafted and closely resemble authentic feedback, they may exert a similar persuasive influence as organic reviews. These “high-quality” manipulated reviews may reinforce consumer perceptions and encourage purchase behavior, particularly when they align with other signals of product quality.

In contrast, poorly written or overly generic manipulated reviews are more likely to be flagged as inauthentic by discerning consumers. In such cases, the perceived manipulation can backfire, undermining trust and decreasing the persuasive power of the reviews. Prior studies suggest that when consumers recognize a review as manipulated, it can increase skepticism, evoke negative reactions, and diminish the review's impact on decision making (Zhuang et al. 2018, Harrison-Walker and Jiang 2023).

Therefore, a key factor shaping the effectiveness of manipulated reviews is the consumer's awareness of review manipulation. The same review may have vastly different effects depending on whether it is accepted as authentic or recognized as deceptive. Our framework thus emphasizes consumer awareness as a critical factor that determines whether manipulated reviews replicate, diverge from, or even reverse the effects typically associated with organic reviews.

This conceptual background provides the foundation for understanding how manipulated reviews influence consumer behavior and, ultimately, product outcomes like app sales. By highlighting the interplay of volume, valence, and consumer awareness, we set the stage to investigate how these elements function

Table 1. Relevant Literature on Review Manipulation

Study	Research focus	Dependent variable	Context	Manipulated review identification strategy	Major findings
Mayzlin et al. (2014)	Firms' incentives for review manipulation	Number of manipulated reviews	Online travel review platform	Comparison between platforms with different policies (TripAdvisor.com and Expedia.com)	Competition affects hotels' incentives to post promotional reviews.
Luca and Zervas (2016)	Firms' incentives for review manipulation	Number of manipulated reviews	Online restaurant review platform	Reviews filtered by Yelp.com	Restaurants with weak reputation or high competition likely engage in review fraud.
Nie et al. (2022)	Firms' incentives for review manipulation	Number of manipulated reviews	Online travel review platform	Comparison between platforms with different policies (TripAdvisor.com and Expedia.com)	Hotels reduce demoting when facing common competitors.
Mo and Li (2018)	Firms' strategies to encourage online reviews	Product rating	Online brand community	N/A	Product sampling increases online ratings from free sample recipients but leads to lower ratings from regular reviewers.
Lin et al. (2019)	Firms' strategies to encourage online reviews	Product rating	E-commerce	N/A	Free product sampling increases product ratings, leading to biased ratings.
Wu et al. (2023)	Firms' strategies to encourage online reviews	Number of manipulated reviews	Online review platform	N/A	Praise cashback elicits very few manipulated reviews.
Gao et al. (2024)	Firms' strategies to encourage online reviews	Review volume	Online travel review platform	N/A	Review solicitations increase both review volume and ratings but crowd out organic reviews.
Lappas et al. (2016)	Review manipulation	Online visibility	Online travel review platform	Simulation	Positive manipulated reviews increase visibility whereas negative manipulated reviews decrease it.
Ananthakrishnan et al. (2020)	Platform governance on review manipulation	Consumer reaction to displayed manipulated reviews	Online restaurant review platform	N/A	Displaying manipulated reviews can increase consumers' trust in the platform.
Zhao et al. (2022)	Platform governance on review manipulation	Volume and valence of online review	Online restaurant review platform	N/A	Displaying filtered reviews discourages users from investing effort in writing reviews.
Kokkodis et al. (2022)	Platform governance on review manipulation	Number of manipulated reviews	Online shopping platform	Reviews filtered by Amazon.com	Optional purchase verification decreases manipulated reviews.
This study	Effects of review manipulation on sales ranking	Sales ranking	Mobile app store	Review filtered by Apple App Store and a large language model	Both positive and negative manipulated reviews have positive effects on apps' sales rankings.

within consumer decision making to better understand the conditions under which manipulated reviews can either benefit or harm sales outcomes.

3.2. Impact of Manipulated Reviews

The impact of manipulated reviews on app sales—driven by their volume and valence—hinges on the

extent to which consumers are aware of review manipulation through processing review information during decision making. We construct a mechanism roadmap (see Figure 1) that illustrates how manipulated reviews affect app sales, depending on whether consumers recognize the reviews as manipulated. Our proposed mechanism outlines four distinct pathways—two for positive manipulated reviews and two for negative manipulated reviews. When manipulation goes unnoticed, the effects mirror those of organic reviews—positive reviews increase sales (scenario PM1), whereas negative ones exert a negative valence effect and a positive volume effect, leaving the net impact undetermined (scenario NM1). When consumers recognize manipulation, the valence effect may backfire: for positive reviews (scenario PM2), perceived inauthenticity can reduce persuasive impact, making the overall effect depend on whether volume can compensate; for negative reviews (scenario NM2), recognizing the manipulation can reduce the deterrent impact of negative content, allowing the volume effect to drive a potential sales boost. We elaborate on the conceptual framework in the following section.

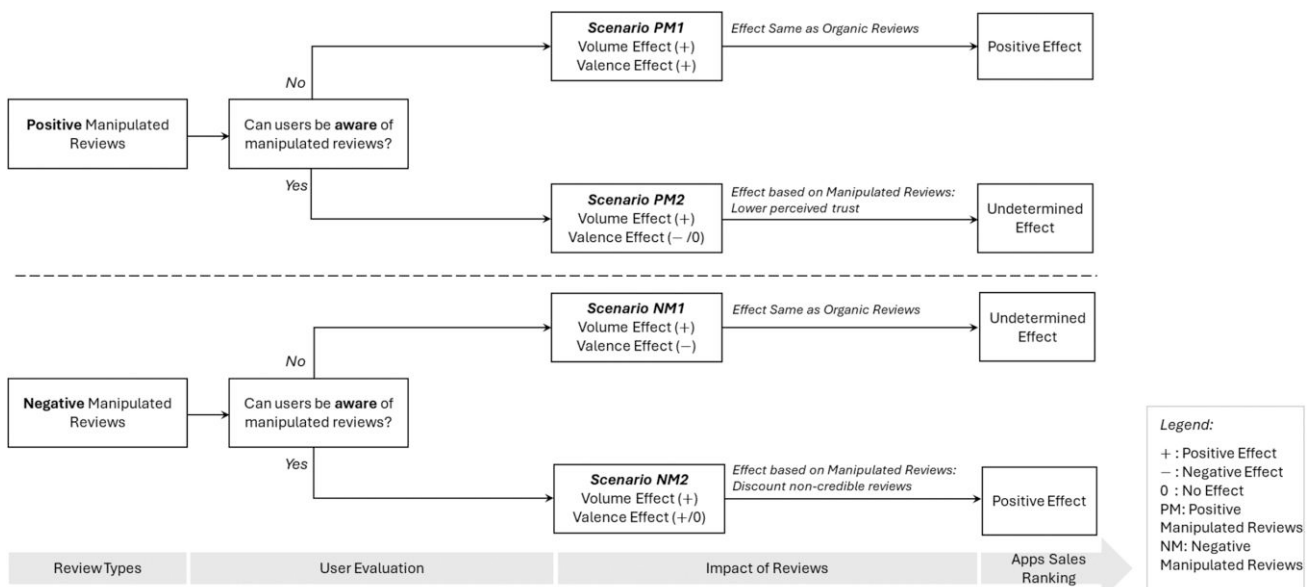
3.2.1. When Consumers Are Not Aware of Review Manipulation. If consumers cannot distinguish manipulated reviews from organic ones, the valence effects mirror those of organic reviews (scenario PM1 and scenario NM1). In the case of positive manipulated reviews, they can enhance a product’s visibility by signaling widespread approval, reinforcing perceptions of quality, and fostering trust and enthusiasm among consumers. This, in turn, increases the app’s perceived value and appeal, ultimately boosting sales. The

combination of increased review volume and positive tone amplifies this effect. Under scenario PM1, we therefore expect positive manipulated reviews to have a favorable impact on product sales.

For negative manipulated reviews, if consumers are not aware of the manipulation, these reviews produce effects similar to their organic counterparts, generating a negative valence effect alongside a positive volume effect. The negative valence effect arises from the unfavorable perceptions these reviews foster. Prior research shows that negative information tends to have a strong impact on consumer attitudes and decision-making processes (Rozin and Royzman 2001, Vaish et al. 2008). As a result, such reviews can depress sales by shaping unfavorable consumer evaluations. For instance, a believable negative manipulated review—such as “The app crashes every time I save my progress, even after updates”—provides specific and relatable details that align with expectations of authenticity, thereby intensifying the valence effect and discouraging purchases. On the other hand, negative manipulated reviews also generate a positive volume effect, which can enhance product visibility and convey social proof. Therefore, under scenario NM1, when consumers are unaware of the manipulation, the overall impact on sales remains undetermined and depends on the relative strength of the valence and volume effects.

3.2.2. When Consumers Are Aware of Review Manipulation. When consumers recognize manipulated reviews, the volume effect may persist, but the valence effect can diverge from that of their organic counterparts. For positive manipulated reviews, research indicates that spotting

Figure 1. Mechanism Roadmap: Effect of Manipulated Reviews on App Sales Ranking



manipulation sparks suspicion, undermining trust and causing reviews to backfire (Petty and Cacioppo 1986, Smith and Vogt 1995, Wee et al. 1995, Ziegele and Weber 2015). Although these reviews may help generate a positive volume effect, their low perceived credibility could dampen their valence impact on user behavior (Smith and Vogt 1995, Ziegele and Weber 2015), leaving their overall impact undetermined (scenario PM2).

On the other hand, negative manipulated reviews that are poorly executed and obvious—such as a vague comment with no depth, “This app is garbage,” or blatant giveaways like “I was paid to write this bad review” or “I haven’t used it, but it’s awful”—fail to convince discerning readers. Consumers who detect these signs could discount such reviews as unreliable or uninformative (Keller and Staelin 1987, Qiu et al. 2017). As a result, the valence effect may diminish, allowing the volume effect—stemming from the presence of more reviews—to play a more prominent role, which can still help sustain visibility and potentially support sales despite the negative content (scenario NM2).

In summary, although both manipulated and organic reviews can generate similar volume effects by enhancing visibility and social proof, their valence effects are far more context-dependent. Specifically, the influence of manipulated reviews on consumer perceptions largely hinges on whether consumers recognize the manipulation. These considerations set the stage for our investigation into how consumer awareness of review manipulation interacts with the decision-making process, shedding light on the conditions under which manipulated reviews may either boost or undermine sales outcomes. Given that our conceptual framework highlights potentially conflicting impacts of positive versus negative manipulated reviews on an app’s sales ranking, we proceed to empirically test these effects.

4. Institutional Setting

The Apple App Store is among the most popular mobile app vendors worldwide, offering almost five million apps (Statista 2024). In the second quarter of 2024, the App Store earned \$24.6 billion, double Google Play’s global revenue of \$11.2 billion (Ceci 2024). Unlike Android devices, Apple’s iOS devices, such as iPhones and iPads, only support app installation via the App Store. Consequently, along with its app reviews and app rankings, the App Store is of ultimate importance to app users and developers. This is not the case for many other contexts, such as Android apps and Amazon merchandise, as those products are available on many markets, making it difficult to see the entire picture of product reviews and sales.

Furthermore, Apple is well-known for exercising exclusive control over its App Store, a factor that is essential for ensuring that it is a reliable source for

iOS app downloads. This strict oversight is also evident in Apple’s app-review management. For instance, creating accounts in the App Store to write reviews is more challenging and time-consuming than it is on Android platforms such as Google Play (Fernandez 2020). In addition, Apple imposes more restrictive app regulations and exercises tighter control over app quality than Android platforms. For purposes of this study, such a level of control can help us avoid unnecessary noise, such as malicious apps.

4.1. App Reviews

The App Store allows consumers to post reviews for the apps they have downloaded. Each app review includes a rating that ranges from one star (for the most negative reviews) to five stars (for the most positive reviews) and a review text with a title and a body. App reviews reflect user perceptions of an app’s quality and provide insights into user acceptance and satisfaction. Figure WA1 in the Web Appendix shows an example of an app’s product page and review page in the App Store. In addition to individual reviews, the App Store highlights several key review metrics on an app’s review page, including the total number of ratings, the average rating, and the star distribution. The star distribution shows consumers the relative volume of negative and positive reviews, offering a clearer picture of user satisfaction and range of opinions. These metrics are prominently displayed at the top of the app’s review page, making them highly visible and hard to ignore.

When shopping for mobile apps to gauge their quality and credibility, most users consider consumer reviews to be essential for making download decisions. Previous research has shown that 77% of app users read at least one review before downloading a free app, and the number rises to 80% for paid apps (Patrathiranond 2020). Furthermore, consumers associate apps that draw lower ratings with lower quality, which weakens their purchase intentions. Research has indicated that 85% of mobile users would not download an app with an average rating of one or two stars, and over 50% would not consider an app with a three-star average rating and lower (Patrathiranond 2020). These figures confirm that app reviews and ratings are indeed critical for consumers’ product evaluation and decision making.

4.2. Review Filtering

Despite its strict regulations, the App Store suffers from severe review manipulation, a serious challenge for many platforms. To tackle this problem, Apple implemented a sophisticated approach that combines machine learning, artificial intelligence, and human inspection by expert teams to maintain the credibility of the reviews posted in the App Store. In addition to

manipulated reviews, Apple also considers reviews that have useless or offensive content to be policy-violating, which closely resembles policy-violating reviews defined by other digital platforms.³ Table WA1 in the Web Appendix presents examples of the Apple App Store's filtered reviews in our data set.

The App Store adopts the postfilter strategy, identifying and removing policy-violating reviews after they become visible to other App Store users. For example, Apple removed 94 million posted reviews from the App Store in 2021 (Apple 2022). However, it often takes Apple about a month to detect and remove suspicious reviews after they appear on the site (Miller 2021), during which time consumers are still able to view them. Typical of such technologies, the detection algorithm developed by Apple is not perfect and may thus filter out appropriate reviews while retaining policy-violating reviews. We also collected extra data from the App Store to further explore how soon Apple removed the policy-violating reviews. The empirical evidence from the data is consistent with the industry knowledge, where a substantial number of policy-violating reviews might stay on the platform for about one month before they are removed by Apple.⁴

4.3. App Ranking

The rapid development of mobile devices has resulted in a massive increase in the number of apps in the App Store, reaching nearly five million apps as of July 2023 (Statista 2024). To alleviate information overload and help consumers navigate a sea of apps, the App Store ranks the most popular apps in charts and divides them into two tabs: Games, which contains all gaming apps (e.g., Angry Birds), and Apps, which contains all nongaming apps (e.g., Google Maps). In each tab, the App Store presents two separate ranking charts based on whether the app is free or paid. In total, there are four ranking charts in the App Store: Free Games, Paid Games, Free Apps, and Paid Apps. Whereas consumers may view the top 200 apps on each ranking chart, Apple allows developers to see the top 1,500 apps on each chart.

5. Empirical Analysis

5.1. Data Sample

We acquired a unique data set that includes every mobile app to appear at least once among the top 1,500 apps in the App Store in China between November 1, 2018, and October 31, 2019. Overall, there are 40,570 apps in our data set, which received over 58 million reviews during the one-year data period. Although all these reviews were visible to consumers when initially posted in the App Store, approximately 10% were later filtered out by Apple. For each app review, we observe the post date, rating level, text content, and whether it

was eventually filtered or not. We refer to reviews later filtered by Apple as "filtered reviews" and other reviews as "published reviews."

In addition to app reviews, we observe each app's ranking, characteristics, featured listing, and keyword management in the App Store on a daily basis. This allows us to construct a data panel at the app and date levels, with a total number of 1,822,342 observations. The data panel has an unbalanced structure, as each app appears on the ranking charts on different days. When an app is not among the 1,500 apps on a ranking chart, the App Store does not assign it a ranking value. As a result, our panel data include all calendar days for an app when it is among the top 1,500 but exclude any days for the app if it is not among the top 1,500. On average, an app appears on the ranking charts in the App Store for 45 days during the one-year data period, with a range from 1 day to 365 days.

5.2. Classification of Filtered Reviews Based on Text Analysis

We leverage a supervised machine learning approach, Bidirectional Encoder Representations from Transformers (BERT), to classify filtered reviews into three nonoverlapping types: (1) manipulated reviews, (2) reviews with useless content that do not address consumer experiences (e.g., a person's political, social, or religious commentary), and (3) reviews with offensive content (e.g., violence or intimidation). To implement this, we first randomly selected 14,328 nonduplicated pieces of reviews from all reviews, and with each piece being human-annotated with one of the three review types. We then randomly allocated 13,808 reviews to the training set, whereas the remaining 520 were assigned to the test set for performance evaluation.

BERT (Devlin et al. 2018) is one of the most advanced large language models with multiple layers of transformers. It has demonstrated strong performance across various natural language understanding tasks, such as text classification and question answering. It has been widely shown that pretrained BERT can be easily fine-tuned with an additional output layer, making it a potential universal text embedding model (Ma et al. 2019). The base model we employed is Bert-Base-Chinese, pretrained on the entire Chinese Wikipedia raw text from the open-source machine learning community Hugging Face.⁵ Our customization involves adding a classification layer to the BERT model, freezing the original BERT parameters, and training the model using our annotated data. This approach streamlines the tuning process to parameters within the classification layer, preserving the universal representation ability of the original BERT model while aligning it with our specific task.

Our model is trained by minimizing cross-entropy loss between the output probability distribution and

Table 2. Performance of Review Classification Model on the Test Set

	Manipulated reviews	Reviews with useless content	Reviews with offensive content
Obs	463	41	16
Precision	0.971	0.933	1.000
Recall	0.996	0.683	0.938
F1-score	0.983	0.789	0.968
Overall accuracy	0.969		

Note. Total observations (Obs) = 520.

the human-annotated review types. The loss is optimized using the Stochastic Gradient Descent optimizer, iterating the training set over 20 epochs. We set the learning rate at 2×10^{-4} and keep all other parameters at their default values. Model evaluation occurs after each epoch on the test set, and the model with the highest accuracy is saved. The entire model is developed based on a leading deep learning framework, PyTorch, along with a machine learning package, Transformers.⁶ The model training and testing are executed on a single GeForce RTX 3070 GPU.

Table 2 summarizes our model's performance in predicting the types of filtered reviews. Given the imbalanced fractions of the three types—manipulated reviews, reviews with useless content, and reviews with offensive content—we employ precision, recall, and F1-score metrics for evaluation. The model achieves a prediction accuracy of 0.969 across the three types, with a detailed performance report for each type included. Results indicate the model's commendable proficiency in identifying different review types. This trained model is subsequently applied to classify all filtered reviews in our data. Figure 2 depicts the log value of the total numbers of all filtered reviews, manipulated reviews, and other filtered reviews (i.e., reviews with useless or offensive content) received by each app.

Specifically, we find that 86% of filtered reviews are manipulated reviews.

5.3. Variables and Descriptive Statistics

5.3.1. App Rankings. Our dependent variable is the daily ranking of each app. Following the previous literature on mobile apps (Kübler et al. 2018, Liang et al. 2019), we use app rankings to measure the sales performance of mobile apps. Product ranking is widely employed as a sales performance measure not only for mobile apps but also for other products such as digital cameras (Archak et al. 2011) and books (Li and Hitt 2008). The ranking of the same app could vary greatly during the one-year data period: the average range of app ranking is 519, with the largest range being 1,497. We compute the average ranking of each app and display the distribution in Figure 3. Most apps have a relatively low average ranking, ranging between 1,000 and 1,300, whereas a small number of apps retain rather high average rankings.

5.3.2. Manipulated Reviews. We are mostly interested in the manipulated reviews received by an app. Specifically, we use the log value of the number of one-star manipulated reviews received by app i ($\text{NumManip}_{it,1star}$) and the log value of the number of

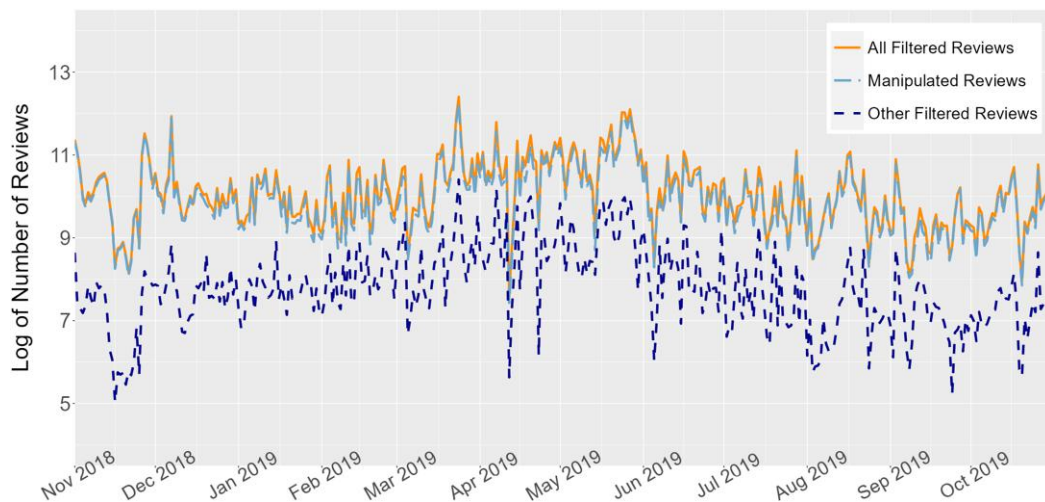
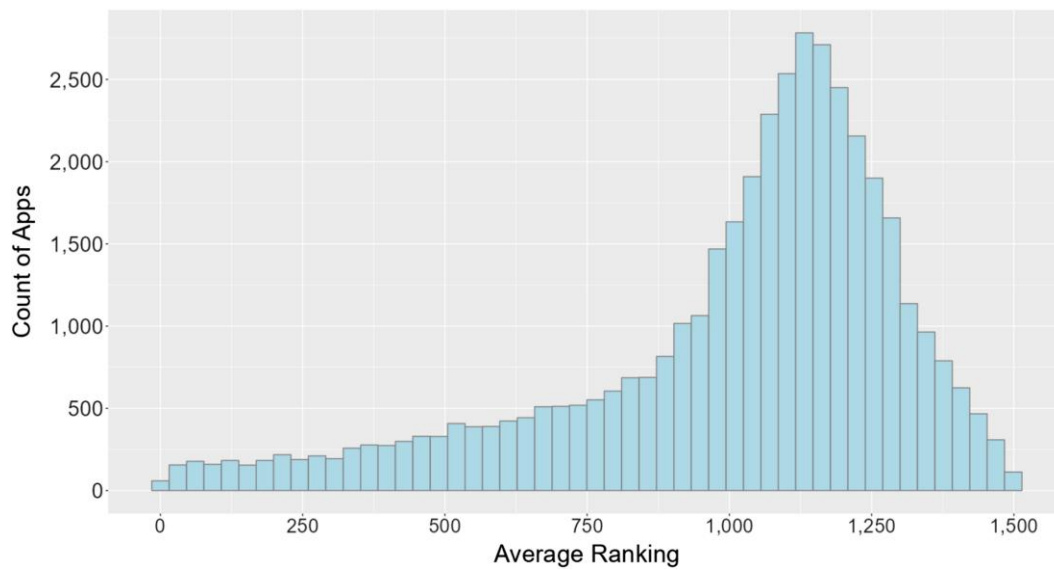
Figure 2. (Color online) Number of Filtered Reviews

Figure 3. (Color online) Distribution of Average Rankings of Apps



five-star manipulated reviews received by app i ($\text{NumManip}_{it,5\text{star}}$). We construct two variables that represent the separate effects of positive and negative manipulated reviews.

We construct the variables for manipulated reviews based on the reviews posted within seven days of day t (i.e., between day $t - 6$ and day t). We choose the seven-day time window because it takes time for Apple to detect manipulated reviews, and recently posted reviews are less likely to be detected and removed by Apple than earlier reviews (Albergotti and Alcantara 2021). Manipulated reviews less than seven days old have a better chance of being viewed by consumers and thus impact app rankings. We also test for alternative time windows (e.g., 1 day and 14 days) in our robustness checks and find consistent results.

Table 3 presents the rating distributions of manipulated reviews. All figures vary greatly across apps, indicated by the large standard deviations. Another emerging pattern is that most manipulated reviews have either one star or five stars, with only a few having two, three, or four stars. This pattern is in line with the findings of Luca and Zervas (2016), showing that manipulated reviews tend to be extreme. Also, there are more one-star manipulated reviews than

five-star manipulated reviews, indicating that negative messages are an attractive strategy for review manipulation.

5.3.3. Control Variables. We control for the effects of published reviews by including $\text{CumNumPublished}_{it}$ and $\text{CumAvgRatingPublished}_{it}$. $\text{CumNumPublished}_{it}$ is the log value of the cumulative number of published reviews received by app i from its release until day t . $\text{CumAvgRatingPublished}_{it}$ is the average rating of cumulative published reviews received by app i during the same period.

We use a dummy variable, $\text{FeaturedListing}_{it}$, to account for whether app i is promoted in featured lists in the App Store on day t . The featured lists are themed collections of apps recommended by Apple’s editorial team, highlighted on the App Store’s main page. When an app is featured in the App Store, it could generate thousands of user impressions, possibly converting to a large number of app downloads (Fishman 2020).

We capture app i ’s keyword management by including a dummy variable, $\text{ActiveKeyword}_{it}$, which takes the value of one if app i uses at least 20 keywords on day t . As the App Store restricts keywords to 100 characters (about 20 keywords in Chinese) (Apple 2022), developers must proactively allocate resources to perform keyword optimization that frequently includes a larger number of accurate keywords. Hence, if app i uses 20 or more keywords on day t , we consider it active in terms of keyword management, because the app fully utilizes the limited keyword space. We also control for the keyword overlap of app i , which is computed as the ratio of the number of common keywords selected by app i —which are also selected by other apps—to the total number of keywords selected by

Table 3. Rating Distributions of Manipulated Reviews

	Mean	SD	Min.	Max.
Number of reviews	215.7	1,551.8	0	69,445
Percentage of 1-star reviews	53.1	46.9	0.0	100
Percentage of 2-star reviews	0.1	0.8	0.0	30.3
Percentage of 3-star reviews	0.2	1.5	0.0	100
Percentage of 4-star reviews	0.2	1.8	0.0	96.9
Percentage of 5-star reviews	46.4	46.6	0.0	100

app i . The more common keywords app i and other apps selected, the more intense the competition app i faces when consumers search for apps using keywords.

Finally, we control for app age in months and average past app ranking. Table 4 reports the descriptive statistics for all variables at the app and date level.

5.4. Model Specification

To estimate the effects of manipulated reviews on app rankings, we consider a fixed effects (FE) panel model with app-specific fixed effects for app i on day t , as follows:

$$y_{it} = \alpha + \beta_1 \cdot \text{NumManip}_{it,1star} + \beta_2 \cdot \text{NumManip}_{it,5star} + \gamma_1 \cdot \text{CumNumPublished}_{it} + \gamma_2 \cdot \text{CumAvgRatingPublished}_{it} + \gamma_3 \cdot x_{it} + \lambda_t + \mu_i + \epsilon_{it}. \quad (1)$$

We log-transform the ranking variable following Chevalier and Mayzlin (2006) and use its negative value: $y_{it} = -\log(\text{app ranking}_{it})$. Hence, the coefficient interpretation becomes straightforward: a positive coefficient indicates that an app's ranking improves when the value of an explanatory variable rises.

We also control for the volume and valence effects of published reviews received by app i . With x_{it} , we account for the featured listing and active keyword management, keyword overlap, app age in months, and negative log of average past app ranking. We include day-of-week fixed effects as well as monthly fixed effects in λ_t to control for unobserved time-related factors such as seasonality and economic trends. We include app-specific fixed effects, μ_i , to control for any unobserved time-invariant app- and developer-related factors. Finally, ϵ_{it} is an idiosyncratic error term and follows a normal distribution: $\epsilon_{it} \sim \mathcal{N}(0, \sigma_\epsilon^2)$. The average variance inflation factor for our model is 1.25, indicating no threat of multicollinearity. The correlation matrix of

all variables is reported in Table WA3 in the Web Appendix.

5.5. Endogeneity of Manipulated Reviews

5.5.1. Endogeneity Issues. Various unobserved factors can correlate with both app rankings and manipulated reviews, potentially causing endogeneity issues and biasing our model's estimates. First, some unobserved factors can be time-invariant characteristics such as app quality, price, and developer reputation. For instance, lower-priced apps tend to attract more downloads compared with higher-priced ones. It is also less expensive for developers to buy manipulated reviews for lower-priced apps. Also, developers with established reputations are less likely to resort to review manipulation for promotion. In contrast, lesser-known developers may rely on review manipulation to boost their app's visibility. Consumers, in general, are also less inclined to trust and download apps from lesser-known developers, preferring those from well-known and trusted sources.

Unobserved factors may also vary across apps and over time. For example, developers might coordinate review manipulation with other marketing campaigns, which can drive app downloads, a phenomenon that is not captured in our data. Other time-variant factors include app updating strategies and competitors' moves. For instance, an app's ranking is likely to fall when its competitors' rankings rise. In this instance, the app has a greater incentive to post positive manipulated reviews to promote itself or negative manipulated reviews to damage its competitors.

5.5.2. Fixed Effects and Instrumental Variables. To address endogeneity concerns, we employ two approaches. First, we follow previous studies on online reviews and use the panel FE model in Equation (1) to account for unobserved app and developer

Table 4. Descriptive Statistics

Variable	Mean	SD	Min.	Max.
Dependent variable				
App ranking	743	422	1	1,500
Manipulated reviews				
Number of 1-star manipulated reviews	7.76	169.24	0	35,567
Number of 5-star manipulated reviews	6.08	192.25	0	48,502
Other control variables				
Cumulative number of published reviews	2,179.16	9,309.78	0	416,858
Cumulative average rating of published reviews	4.08	1.22	0	5
Featured listing	0.06	0.23	0	1
Active keyword	0.63	0.48	0	1
Keyword overlap	0.05	0.06	0	1
App age (months)	35.84	29.97	0.03	137.67
Average past app ranking	730	365	1	1,500
Number of observations		1,822,342		
Number of apps		40,570		

characteristics such as app quality and developer reputation (Chevalier and Mayzlin 2006, Chen et al. 2011). By allowing for arbitrary correlations between the unobserved fixed effects, μ_i , and the independent variables, we can effectively cancel out the fixed effects using a within transformation.

Second, we use instrumental variables to tackle the remaining endogeneity issues related to $\text{NumManip}_{it,1star}$ and $\text{NumManip}_{it,5star}$. To do this, we identify apps with similar motivations and costs for review manipulation as app i and use the average values of their $\text{NumManip}_{it,1star}$ and $\text{NumManip}_{it,5star}$ as instrumental variables. The identified apps must meet the following criteria:

- Condition 1: The identified apps' price, average ranking, and the number of days on ranking charts should be within one standard deviation of app i . If app i is free, only free apps are considered.
- Condition 2: The identified apps' number of one- or five-star manipulated reviews in the last two months should be within 0.5 standard deviation of app i . This tighter range accounts for the higher variability in manipulated reviews compared with the other variables (Table 4).
- Condition 3: The identified apps should belong to different categories and ranking charts, and be developed by different developers than app i . For example, if app i is a paid game, we identify apps, such as paid healthcare apps, and ensure they are developed by different app developers.

We estimate the effects of manipulated reviews on app rankings using a two-stage least squares (2SLS) model with panel data. The first stage includes the instrumental variables, IV_{it} , along with all covariates and fixed effects from Equation (1).

$$\begin{aligned} w_{it} = & \eta_0 + \eta_1 \cdot IV_{it} + \eta_2 \cdot \text{CumNumPublished}_{it} \\ & + \eta_3 \cdot \text{CumAvgRatingPublished}_{it} \\ & + \eta_4 \cdot x_{it} + \lambda_t + \mu_i + e_{it}, \end{aligned} \quad (2)$$

where $w_{it} = (\text{NumManip}_{it,1star}, \text{NumManip}_{it,5star})$ represents the vector of manipulated review variables. We further validate our instruments below from two perspectives: instrument relevance and exclusion restrictions.

5.5.3. Relevance. We expect the identified apps' $\text{NumManip}_{it,1star}$ and $\text{NumManip}_{it,5star}$ to be strongly correlated with those of app i . First, an app's price significantly affects the cost of posting manipulated reviews; before they can leave a review, users need to pay the App Store and developers to download apps that are not free. Second, the average ranking and the number of days on ranking charts can capture the market status of the apps. We expect apps with

similar market status to have similar motivations for review manipulation (Mayzlin et al. 2014, Luca and Zervas 2016). In addition, having a similar number of manipulated reviews in the previous two months ensures the identified apps have similar recent experiences of review manipulation. For example, apps connected to the black market for purchasing manipulated reviews can easily boost their review volume.

To validate the relevance of our instruments, we report the estimation results of the first-stage equations in Table WA4 in Web Appendix B. The significant coefficient estimates demonstrate the strong relevance of our instruments. Furthermore, the partial F statistical tests on the instruments show consistently high values across the first-stage equations, indicating that the instruments effectively account for the variation in the endogenous variables.

5.5.4. Exclusion Restriction. The identified apps are in different categories and ranking charts, and are developed by different developers from app i . As a result, we do not expect the manipulated reviews of these identified apps to directly impact the ranking of app i . For example, if app i is a paid gaming app, we exclude other gaming apps from the identified apps for constructing instruments. We do not anticipate manipulated reviews of apps—such as paid healthcare apps—to influence the ranking or downloads of the gaming app i , especially because they are ranked in different charts and are not developed by the same app developer. Moreover, we control for the same set of control variables, app-specific fixed effects, and time-fixed effects per Equation (1) in the first-stage equation for the instrumental variables. Conditional on these variables, it is unlikely that the manipulated reviews of the identified apps are directly correlated with app i 's ranking.

5.5.5. GMM Approach. As a robustness check, we leverage the dynamic panel nature of our data and employ the generalized method of moments (GMM) estimator to address endogeneity concerns (Arellano and Bond 1991, Blundell and Bond 1998). As detailed in Web Appendix D, we utilize the internal instruments provided by dynamic panel data and the GMM approach, which is increasingly gaining popularity in information systems (IS) research (e.g., Saldanha et al. 2020, 2022; Xue et al. 2021; Jabr and Rahman 2022; Fürstenau et al. 2023; Li et al. 2024).⁷

6. Results and Discussions

6.1. The Main Effects of Manipulated Reviews

In this section, we discuss our main findings based on the estimation results reported in Table 5. Column (1) reports the estimates without instrumental variables,

Table 5. Effects of Manipulated Reviews

	Without instruments (1)	With instruments (2)
NumManip _{1star}	0.046*** (0.004)	0.092*** (0.020)
NumManip _{5star}	0.038*** (0.004)	0.111*** (0.025)
CumNumPublished	−0.012*** (0.003)	−0.022*** (0.004)
AvgRatingPublished	0.012*** (0.003)	0.011*** (0.003)
Featured listing	0.324*** (0.012)	0.320*** (0.012)
Active keyword	0.024** (0.009)	0.024** (0.009)
Keyword overlap	−0.166*** (0.049)	−0.150*** (0.049)
App age	−0.007*** (0.002)	−0.010*** (0.002)
Log of average past ranking	0.941*** (0.010)	0.924*** (0.012)
Constant	−0.130 (0.085)	−0.120 (0.087)
App fixed effects	Yes	Yes
Day-of-week fixed effects	Yes	Yes
Month fixed effects	Yes	Yes
R ²	0.725	0.669
F statistics	449.62	—
Wald χ^2	—	115,685.33
Observations	1,822,342	1,822,342
Number of apps	40,570	40,570

Note. Robust standard errors are reported in parentheses.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

whereas column (2) provides the estimates with instrumental variables. Although the coefficients for the manipulated reviews in column (1) align qualitatively with those from the model with instrumental variables, there are notable differences in the magnitudes of the results. This highlights the importance of addressing the issue of endogeneity.

The results indicate that five-star manipulated reviews have a positive and statistically significant effect on app rankings. Specifically, a 100% increase in the number of five-star manipulated reviews is related to an 11.1% increase in an app's ranking. Surprisingly, a 100% increase in the number of one-star manipulated reviews correlates with a 9.2% increase in an app's ranking. Given that the average app ranking in our data set is 743, this effect translates into an improvement of approximately 82 and 68 positions, respectively.

These findings suggest that both scenarios PM1 and PM2 are plausible for five-star manipulated reviews, whereas both scenarios NM1 and NM2 are plausible for one-star manipulated reviews. In order to investigate which scenarios are more plausible, we conducted a text analysis to examine the extent to which consumers are able to recognize positive versus

negative manipulated reviews. Specifically, we examine how detectably different the manipulated reviews are from their organic counterparts, by comparing the linguistic similarity between manipulated and organic reviews within each app.

We first represented each review using a bag-of-words model with term frequency-inverse document frequency (TF-IDF) weighting (Aizawa 2003, Ramos 2003, Qaiser and Ali 2018), transforming the text into a numerical vector. To assess linguistic similarity, we grouped reviews within each app into four clusters based on their rating and authenticity: one-star manipulated, one-star organic, five-star manipulated, and five-star organic reviews. For each cluster, we computed a centroid, defined as the average of all TF-IDF vectors within the group. We then measured the cosine similarity between the centroids of manipulated and organic reviews within each rating category (i.e., between one-star manipulated and one-star organic, and between five-star manipulated and five-star organic) for each app. The similarity score ranges between zero and one, with a higher similarity score indicating that the linguistic features of manipulated reviews more closely resemble those of organic reviews, thereby making them harder to distinguish.

The results show that the average similarity among five-star reviews is 0.810 ($SE = 0.008$), compared with 0.578 ($SE = 0.010$) among one-star reviews. The high similarity in the five-star category suggests that manipulated positive reviews closely resemble their organic counterparts, making it more difficult for consumers to discern between organic and manipulated reviews. This increases the likelihood of scenario PM1, in which consumers are unable to detect manipulation, allowing the manipulated reviews to positively impact sales. In contrast, manipulated negative reviews are more linguistically distinct from organic ones, making consumers more likely to identify them as inauthentic. This increases the likelihood of scenario NM2, where recognition of manipulation mitigates the negative impact. Taken together, the text analysis suggests an asymmetry in review detectability and offers an explanation for the observed positive effects of both positive and negative manipulated reviews on app rankings.

6.2. Heterogeneous Effects of Manipulated Reviews

Our main analysis suggests that manipulated reviews—whether positive or negative—tend to have a positive impact on app sales. This finding underscores the potential for manipulated reviews to mislead consumers and distort their choices. However, this average effect may obscure important variations across contexts. In this section, we investigate the heterogeneous effects of manipulated reviews by identifying situations in which their influence may be diminished.

We anticipate that when consumers face higher stakes—for instance, when the app involves a financial cost, supports important tasks (e.g., finance, productivity, or education), or comes from a less-established developer—they are likely to make decisions more cautiously. In such cases, consumers may take additional steps—such as seeking information beyond reading online reviews—to reduce risk or uncertainty before downloading. As a result, we expect the impact of manipulated reviews to be weaker in high-stakes or high-scrutiny contexts, where consumers are less likely to rely solely on online reviews.

First, we examine whether the effects of manipulated reviews differ between free and paid apps. Unlike free apps, paid apps require consumers to incur a financial cost, which raises the stakes of the decision and encourages more deliberate evaluation. The presence of financial stakes encourages consumers to approach the download more cautiously, which in turn diminishes the misleading influence of manipulated reviews. Consistent with this expectation, columns (1) and (2) of Table 6 show that both one- and five-star manipulated reviews have significant positive effects on the sales rankings of free apps, but their effects are weaker and statistically insignificant for paid apps.

Columns (3) and (4) of Table 6 explore differences by app function—specifically, comparing gaming and

nongaming apps. Nongaming apps, such as those related to finance, productivity, or education, tend to serve practical or utilitarian purposes. Because these apps are often tied to important tasks or long-term use, consumers face higher perceived stakes when choosing to download them and are therefore more likely to approach the decision with caution. In contrast, gaming apps are primarily used for entertainment and are typically downloaded for short-term use or experimentation. These decisions tend to be more impulsive and guided by surface-level cues—such as visuals, popularity, or trends—rather than careful deliberation (Arnold and Reynolds 2003, Herabadi et al. 2009). The lower perceived cost of making a poor choice further reduces the motivation to scrutinize such decisions. As a result, manipulated reviews tend to have a stronger influence on gaming apps, where consumers are less cautious. The results indicate that positive manipulated reviews increase sales rankings for both gaming and nongaming apps, with a stronger effect observed for gaming apps. For negative manipulated reviews, the effect is positive and significant for gaming apps but not statistically significant for nongaming apps.

Lastly, we explore heterogeneity by developer size, comparing apps from large versus small developers. Large developers are generally more established and perceived as more trustworthy, which may lead

Table 6. Heterogeneous Effects of Manipulated Reviews

	Free apps (1)	Paid apps (2)	Gaming apps (3)	Nongaming apps (4)	Large developer (5)	Small developer (6)
NumManip _{1star}	0.125*** (0.022)	0.082 (0.121)	0.051** (0.025)	0.037* (0.021)	0.123*** (0.040)	0.068*** (0.020)
NumManip _{5star}	0.154*** (0.028)	0.080 (0.113)	0.076*** (0.027)	0.010 (0.027)	0.127*** (0.049)	0.092*** (0.027)
CumNumPublished	−0.028*** (0.006)	0.010* (0.005)	−0.022*** (0.005)	−0.003 (0.004)	−0.034*** (0.007)	−0.013*** (0.005)
AvgRatingPublished	0.022*** (0.006)	−0.003 (0.004)	0.016*** (0.004)	0.008* (0.005)	0.009* (0.005)	0.012*** (0.004)
Featured listing	0.280*** (0.016)	0.369*** (0.018)	0.380*** (0.014)	0.196*** (0.019)	0.339*** (0.016)	0.285*** (0.019)
Active keyword	−0.013 (0.037)	0.016 (0.010)	0.032*** (0.011)	0.025 (0.016)	0.038*** (0.013)	0.012 (0.013)
Keyword overlap	−0.164* (0.100)	−0.095** (0.048)	−0.266*** (0.061)	−0.089 (0.073)	−0.195** (0.079)	−0.123** (0.059)
App age	−0.032*** (0.004)	0.002 (0.002)	−0.012*** (0.003)	−0.004 (0.003)	−0.013*** (0.004)	−0.007*** (0.003)
Log of average past ranking	0.896*** (0.016)	0.939*** (0.020)	0.908*** (0.015)	0.982*** (0.018)	0.907*** (0.019)	0.939*** (0.015)
Constant	0.278** (0.122)	−0.451*** (0.142)	−0.244** (0.102)	0.035 (0.142)	−0.090 (0.132)	−0.150 (0.111)
App fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Day-of-week fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	949,058	873,284	907,836	914,506	884,884	937,458
Number of apps	17,577	22,993	15,406	25,164	16,796	23,774

Note. Robust standard errors are reported in parentheses.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

consumers to rely more heavily on simple cues such as review volume when making decisions. In contrast, apps from smaller or lesser-known developers may raise greater privacy or security concerns, prompting consumers to approach these downloads more cautiously. As a result, we expect manipulated reviews to exert a stronger influence on apps from large developers and a weaker influence on those from smaller ones. In our data set, developers with up to two apps are classified as small, whereas those with three or more are considered large, representing 58.6% and 41.4% of the sample, respectively. Estimation results in columns (5) and (6) of Table 6 confirm this pattern: both positive and negative manipulated reviews have a stronger impact on apps from large developers than those from small developers.

Taken together, these results highlight the conditions under which the influence of manipulated reviews is more or less pronounced. Across all three dimensions—app price, function, and developer size—we observe a consistent pattern: the misleading impact of manipulated reviews is weaker in settings where consumers are likely to make more cautious decisions. This attenuation is particularly evident when downloads involve financial cost (paid apps), serve utilitarian purposes (nongaming apps), or come from less-familiar developers (small developers)—contexts that raise the stakes of the decision and reduce consumers' susceptibility to misleading signals.

6.3. Reviews with Useless and Offensive Content

Our text analysis classifies filtered reviews into three types, prompting an exploration of their effects based on review content. We first examine the effects of reviews with useless content and observe significant positive effects for both one- and five-star reviews, as shown in column (1) of Table 7. For reviews with offensive content, we also find significant positive effects in column (2), but these effects are much larger compared with those of manipulated reviews or reviews with useless content. The findings consistently show that the volume effect of negative filtered reviews outweighs their valence effects, which strengthens our confidence in the main effects while providing more nuanced insights.

6.4. Long-Term Effects of Manipulated Reviews

Another important aspect to consider is how the effects of manipulated reviews evolve over time, particularly after the platform filters these reviews. We estimate the lagged effects of manipulated review variables that are lagged by one to six months using the fixed effects model with instrumental variables in Equation (1), as reported in Table 8. Although our main analysis shows that the manipulated reviews initially have a positive effect on app rankings, their

Table 7. Effects of Other Types of Filtered Reviews

	Useless content (1)	Offensive content (2)
NumUseless _{1star}	0.211*** (0.046)	— —
NumUseless _{5star}	0.269*** (0.061)	— —
NumOffensive _{1star}	— —	1.059*** (0.285)
NumOffensive _{5star}	— —	2.398*** (0.817)
CumNumPublished	−0.017*** (0.003)	−0.017*** (0.004)
AvgRatingPublished	0.013*** (0.003)	0.027*** (0.005)
Featured listing	0.324*** (0.012)	0.321*** (0.013)
Active keyword	0.025** (0.010)	0.023* (0.014)
Keyword overlap	−0.151*** (0.050)	−0.102 (0.067)
App age	−0.011*** (0.002)	−0.009*** (0.003)
Log of average past ranking	0.921*** (0.012)	0.859*** (0.029)
Constant	−0.121 (0.087)	−0.647*** (0.177)
App fixed effects	Yes	Yes
Day-of-week fixed effects	Yes	Yes
Month fixed effects	Yes	Yes
Observations	1,822,342	1,822,342
Number of apps	40,570	40,570

Note. Robust standard errors are reported in parentheses.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

effects turn negative after about one month of posting. Notably, five-star manipulated reviews begin to show significant negative effects sooner than one-star manipulated reviews. Over time, the negative effects of one-star manipulated reviews have also become significant.

This pattern supports our earlier discussion: manipulated reviews are not immediately filtered out after being posted in the App Store; it may take one month for the App Store to effectively detect and remove them. Also, as these manipulated reviews are filtered out, their influence on rankings diminishes, and the initial positive effects fade away. Furthermore, apps with more previous manipulated reviews were more likely to experience subsequent filtering, which could then negatively impact their rankings and mitigate the initial distortion effects caused by manipulated reviews. According to our analysis of long-term effects, such mitigation could take as long as six months to fully materialize.

6.5. Robustness Checks

We conduct additional analyses to ensure the robustness of our findings, reported in Web Appendix E.

Table 8. Long-Term Effects of Manipulated Reviews

	Manipulated reviews posted N months ago					
	$N = 1$ (1)	$N = 2$ (2)	$N = 3$ (3)	$N = 4$ (4)	$N = 5$ (5)	$N = 6$ (6)
NumManip _{1star}	−0.011 (0.012)	−0.039*** (0.013)	−0.045*** (0.014)	−0.043*** (0.014)	−0.048*** (0.015)	−0.013 (0.013)
NumManip _{5star}	−0.047** (0.018)	−0.060*** (0.019)	−0.025 (0.021)	−0.054** (0.022)	−0.043* (0.024)	−0.022 (0.021)
CumNumPublished	−0.001 (0.004)	0.001 (0.005)	−0.005 (0.005)	−0.004 (0.006)	−0.010 (0.007)	−0.013* (0.008)
AvgRatingPublished	0.013*** (0.004)	0.013*** (0.004)	0.018*** (0.005)	0.018*** (0.005)	0.018*** (0.005)	0.018*** (0.006)
Featured listing	0.336*** (0.012)	0.335*** (0.013)	0.333*** (0.013)	0.325*** (0.013)	0.324*** (0.013)	0.326*** (0.014)
Active keyword	0.027*** (0.010)	0.032*** (0.012)	0.026** (0.013)	0.016 (0.015)	0.011 (0.016)	0.016 (0.018)
Keyword overlap	−0.203*** (0.055)	−0.197*** (0.059)	−0.214*** (0.067)	−0.201*** (0.075)	−0.205** (0.083)	−0.258*** (0.091)
App age	−0.007*** (0.002)	−0.006** (0.002)	−0.012*** (0.002)	−0.010*** (0.003)	−0.011*** (0.003)	−0.015*** (0.003)
Log of average past ranking	0.936*** (0.012)	0.912*** (0.014)	0.904*** (0.015)	0.887*** (0.016)	0.885*** (0.017)	0.890*** (0.017)
Constant	−0.169* (0.098)	−0.352*** (0.118)	−0.207 (0.130)	−0.375*** (0.134)	−0.343** (0.143)	−0.137 (0.144)
App fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Day-of-week fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.713	0.713	0.617	0.642	0.623	0.538
Observations	1,672,470	1,522,562	1,372,800	1,223,205	1,073,447	923,555
Number of apps	38,282	36,132	34,228	32,173	30,222	27,867

Note. Robust standard errors are reported in parentheses.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

First, we estimate the effects of manipulated reviews using the System GMM (SGMM) model, and the results in column (1) of Table WA5 indicate that both effects remain positive. In addition, we use a panel Tobit model to address the issue of the ranking variable that is censored, where we do not observe an app's ranking when it drops out of the ranking chart. However, we observe manipulated reviews as well as other variables on those days, and our estimation results based on a random sample of 3,000 apps are consistent with our main analysis (see column (2) of Table WA5).

Second, for our main analysis, we construct variables for manipulated reviews using a seven-day window, but the effects of manipulated reviews may have shorter or longer lag times depending on how soon the App Store detects the manipulated reviews and how likely consumers are to read them. We therefore test two alternative time windows: same-day (i.e., no lag) and a 14-day lagged window to construct the manipulated review variables. The results, presented in Table WA6 in the Web Appendix, are consistent with our main analysis.

Furthermore, we use the log number of one- and five-star manipulated reviews to capture the effects in our main analysis. We are also interested in six

alternative measures: (1) only including the log number of one-star manipulated reviews; (2) only including the log number of five-star manipulated reviews; (3) the absolute number of one- and five-star filtered reviews scaled by 1,000; (4) applying our main model to a subsample of the panel data, focusing on observations where NumManip_{it,1star} or NumManip_{it,5star} have nonzero values; (5) the log number of one- and five-star filtered reviews; and (6) the log number of negative (one- and two-star) and positive (four- and five-star) manipulated reviews. Our results remain similar in Table WA7 in the Web Appendix.

In addition, our findings are robust to alternative model specifications. As seen in column (1) of Table WA8 in the Web Appendix, we estimate our main model by using the ranking change in the previous seven days as an alternative dependent variable. In column (2) we exclude all control variables, and in column (3) we control for the number of one- and five-star published reviews in the previous seven days of day t instead of cumulative published reviews.

Finally, we estimate the immediate and lagged effects of one-star manipulated reviews in one model and five-star manipulated reviews in another. To address multicollinearity concerns, we use five model

specifications with different instrumental variable strategies. Given the high correlation among fitted values when instrumenting all review variables simultaneously, we report results with instruments applied to one review variable at a time. This approach mitigates multicollinearity and ensures reliable estimates. The results remain consistent with our main analysis, reinforcing the dynamic effects of manipulated reviews over time (see Tables WA9 and WA10 in the Web Appendix).

7. Discussion and Conclusions

This study examines the effects of manipulated reviews on product sales ranking in the context of the Apple App Store, using detailed daily data on app rankings as well as manipulated reviews detected and filtered by Apple. We build a theoretical framework that extends the literature on review volume and valence to the context of manipulated reviews, emphasizing the pivotal role of consumer awareness of manipulated reviews. We find that positive manipulated reviews exert a positive effect on app sales, suggesting consumers might fail to detect manipulation. Although negative manipulated reviews are often designed to damage an app's reputation, our findings suggest they may unintentionally increase app visibility and improve sales performance. This counterintuitive effect likely arises because many consumers could recognize the reviews as fake and disregard them. Notably, our text analysis reveals that negative manipulated reviews are linguistically more distinguishable from organic ones than are positive manipulated reviews, making them easier for consumers to spot. This implies an asymmetry in consumer response: negative manipulated reviews may be more likely to be detected, whereas positive manipulated reviews tend to go unnoticed.

Our heterogeneity analyses further support our theoretical framework's core premise. In contexts where consumers are more likely to evaluate apps more cautiously—such as paid apps, where the financial cost prompts users to assess value more carefully; non-gaming apps, which often serve functional or long-term purposes that encourage thoughtful evaluation; and apps from less-established developers, where unfamiliarity raises concerns about quality or trustworthiness—the influence of manipulated reviews diminishes. These findings suggest that the effectiveness of manipulated reviews depends critically on consumer scrutiny and is shaped by important boundary conditions. These conditions—such as the nature of the app, its price, and the credibility of its source—reflect underlying mechanisms related to perceived risk and motivation to process information.

Finally, our long-term analysis shows that although manipulated reviews may offer short-term gains,

these effects gradually reverse as the platform detects and removes manipulated content. This process can take several months, during which the initial benefits fade and may even turn negative, particularly for positive manipulated reviews. This time lag highlights the critical role of platform governance in mitigating manipulation and restoring trust in review systems.

Our study extends prior research by shifting the focus from the motivations behind review manipulation and platform-level detection strategies to the downstream consequences of manipulated reviews on market outcomes. Whereas existing literature has largely examined why firms engage in manipulation and how platforms respond, we explore how manipulated reviews—both positive and negative—influence app sales rankings. Second, from a methodological perspective, we leverage a large language model to more accurately isolate manipulated reviews from other filtered content, offering a scalable and replicable tool for future detection efforts and enhancing the precision of empirical investigations in this domain. Finally, our theoretical arguments extend the well-documented valence and volume effects observed in organic reviews to the context of manipulated reviews, suggesting that the influence of these mechanisms is shaped by how consumers engage with the manipulated online reviews.

7.1. Managerial Implications

Our research provides not only insights for researchers who study manipulated reviews but also managerial implications for practitioners beyond mobile app markets. Naturally, the issue of manipulated reviews is not limited to the App Store, as it also affects many major platforms such as TripAdvisor, Yelp, Amazon, and Google. Manipulated reviews continue to disrupt major digital platforms such as travel booking hubs and e-commerce sites, causing problems for both consumers and businesses. Our findings thus provide crucial insights for those platforms as well.

Although prompt filtering of manipulated reviews may reduce undesired impacts on a platform, the system is not always effective and usually comes at a high financial cost for human inspections and machine learning methods used for filtering algorithms. Furthermore, users may be irritated if their appropriate or honest reviews are removed instantly by a flawed filtering system. As a result, digital platforms face a trade-off when deciding whether to filter out manipulated reviews in a timely manner and what level of technology and human resources they apply to the process. Yet our findings indicate that manipulated reviews are strongly associated with distortions in product rankings and sales on digital platforms before these reviews are filtered, thus highlighting the need for platforms to quickly and accurately increase their

investment in filtering manipulated reviews to uphold consumer trust and reputation.

In addition, managers of online marketplaces could benefit from our discovery that manipulated reviews have stronger effects on specific types of products, such as apps designed by large developers, free apps, and game apps. In their review detection efforts, online marketplaces could therefore target products that may help them effectively reduce the damage associated with such problematic reviews. This approach could prove especially valuable considering that current penalties for posting manipulated reviews, such as app removal and developer expulsion, occur infrequently and are thus not regarded as a serious risk by app developers (Hill 2018).

7.2. Limitations and Future Research

This study is subject to several limitations that may create opportunities for future research. First, we only observe when the manipulated reviews were posted in the App Store, but not the exact time they were filtered out by Apple. Future studies may investigate whether and how the timing of review filtering moderates the effects of manipulated reviews by collecting the corresponding data. Second, we are unable to observe the number of downloads or revenues of the apps included in our analysis. Instead, we use rankings to measure the market performance of apps. In the future, however, researchers may be able to estimate the effects of manipulated reviews on product sales by obtaining sales or revenue data. Third, we concentrate on the effects of manipulated reviews of an app based on its own sales ranking. Yet the sales ranking of a focal app may also be influenced by manipulated reviews received by its competitors, leading to a potential spillover effect. We leave the exploration of this effect for future research. Finally, our heterogeneity analysis highlights several boundary conditions that shape the effectiveness of manipulated reviews. Building on this foundation, future work could investigate additional contextual factors, such as the sensitivity of the app domain (e.g., health, finance, or children's products), differences in user expertise, or different forms of product differentiation (e.g., horizontal and vertical differentiation), which may further moderate consumer responses. Exploring these conditions would help clarify when manipulated reviews are most likely to influence behavior and when they are more prone to backfiring, especially in high-risk settings where consumers are more alert and discerning.

Endnotes

¹ Sources: <https://support.google.com/contributionpolicy/answer/7400114>, <https://www.yelp.com/guidelines>, and <https://www.airbnb.com/help/article/2673/airbnbs-review-policy>.

² Some reviews are filtered out by digital platforms if they are deemed policy-violating, such as reviews that promote a competing platform. Whereas consumers might perceive some of these reviews as appropriate or helpful, the platform views them as potentially harmful to its reputation or business.

³ Source: <https://www.apple.com/legal/internet-services/itunes/>.

⁴ Please see details in Web Appendix A.

⁵ <https://huggingface.co/bert-base-chinese>.

⁶ <https://pytorch.org/>; <https://huggingface.co/docs/transformers/index>.

⁷ Notably, Cheng and Bang (2019) provide a comprehensive summary of research papers published in major IS journals (*Information Systems Research*, *Journal of Management Information Systems*, and *MIS Quarterly*), where 14 of these papers utilize GMM estimation as their main analysis method, whereas an additional 19 papers employ GMM estimation for robustness checks.

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